Blood Bank Inventory & Emergency Donor Matcher

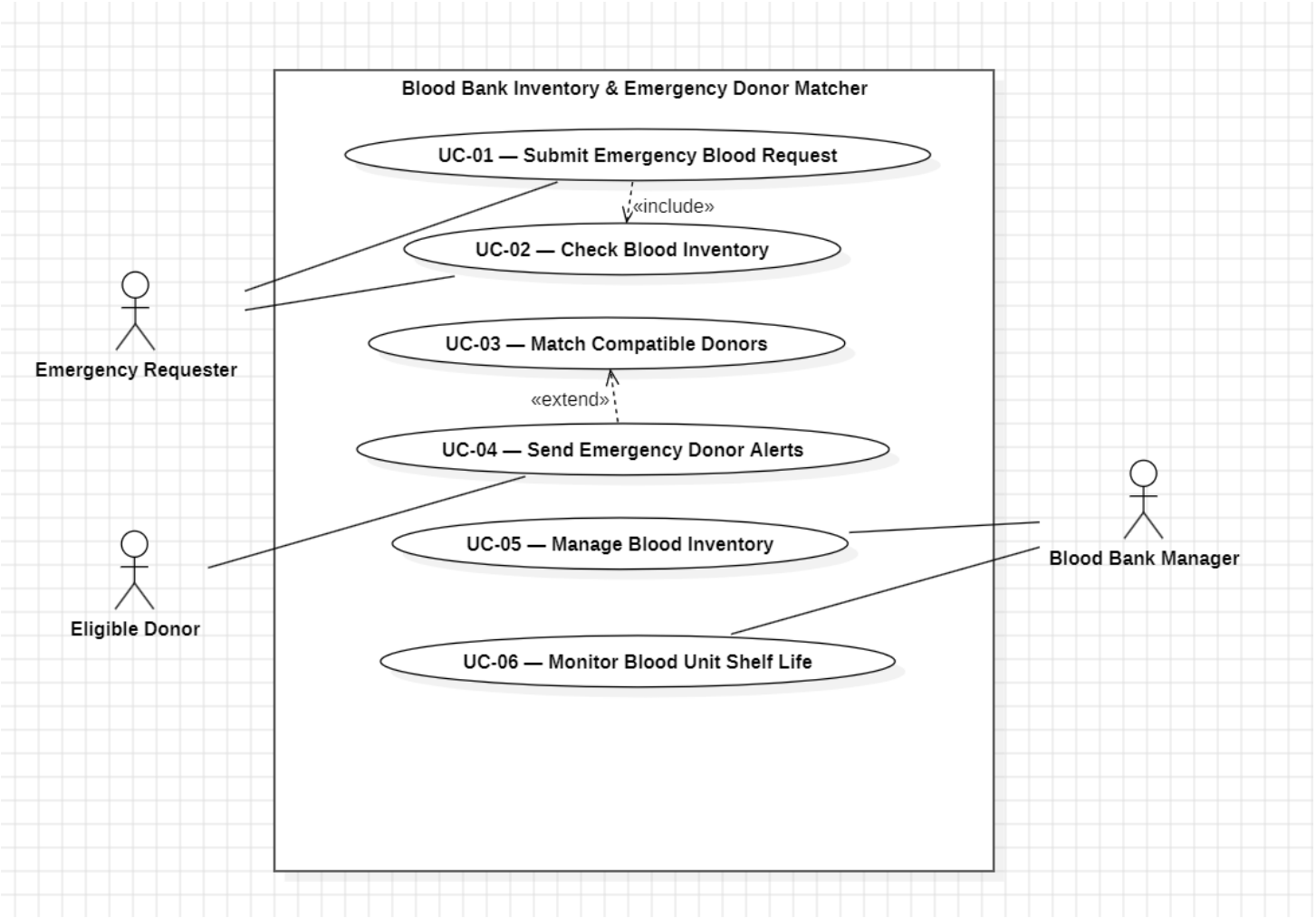
Lab 1 – Requirements Engineering & UML Use-Case Modelling

1. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (short)	Comments
FR-001	Functional	The system shall cross-reference emergency blood unit requests against real-time blood bank inventory and broadcast SMS alerts to compatible donors within a 10 km radius.	High	PASS: For a valid emergency request, the system checks current inventory and, when donor support is required, sends SMS alerts only to compatible donors within 10 km. FAIL: the request is not checked against current inventory or an incompatible/out-of-range donor is alerted.	Core emergency matching and notification function specified in the problem statement.	Clear and easy to test.
FR-002	Functional	The system shall allow the Blood Bank Manager to add, update, and remove blood inventory records for blood components and quantities.	High	PASS: An authorized manager can create, update, and remove an inventory record and the changed quantity is displayed after the operation. FAIL: an unauthorized or invalid operation changes the inventory.	Supports accurate real-time inventory management.	Specify who can update inventory.
FR-003	Functional	The system shall allow the Blood Bank Manager to monitor blood unit shelf life and identify units approaching expiry.	Medium	PASS: For every stored blood unit, the system displays its expiry information and flags units meeting the configured approaching-expiry condition. FAIL: an approaching-expiry unit is not flagged.	Helps the blood bank monitor usable stock and shelf life.	Define the expiry warning period
FR-004	Functional	The system shall verify donor compatibility and eligibility before including a donor in an emergency notification list.	High	PASS: Only donors satisfying the required compatibility and eligibility conditions appear in the notification list. FAIL: any incompatible or ineligible donor appears in the list.	Prevents unsuitable donors from being contacted during an emergency.	Looks clear.
FR-005	Functional	The system shall record the status of each emergency blood request and its associated donor notification activity.	Medium	PASS: After processing an emergency request, the system displays a retrievable status showing the request outcome and notification status. FAIL: the request or notification status cannot be retrieved.	Provides traceability for emergency requests and donor outreach.	Specify what request statuses are stored.
NFR-001	Nonfunctional	The inventory ledger must maintain strict transactional consistency, preventing simultaneous allocation of the same blood bag.	High	PASS: Concurrent allocation tests confirm that one blood bag cannot be successfully allocated to two requests. FAIL: the same blood bag is allocated to multiple requests.	Protects inventory integrity during simultaneous emergency requests.	Clear and testable.
NFR-002	Nonfunctional	The system shall restrict access to blood inventory and donor information to authorized system users according to their assigned role.	High	PASS: An authorized user can access permitted information, while an unauthorized user is denied access in access-control tests. FAIL: an unauthorized user can view or modify restricted information.	Protects sensitive blood bank and donor information from unauthorized access.	Specify access for each role.

2. UML Use-Case Diagram

Final StarUML diagram



3. Use-Case Flow Document

Use Case: Submit Emergency Blood Request

Use Case	Submit Emergency Blood Request
Primary Actor	Emergency Requester
Supporting Actor(s)	Blood Bank Manager; Eligible Donor

Main Success Scenario

Step	Main Success Scenario
1	Emergency Requester selects “Submit Emergency Blood Request.” The system displays the request form.
2	Emergency Requester enters the required blood group, blood component, quantity, and urgency.
3	The system validates the entered request details.
4	The system records the emergency blood request.
5	The system checks the request against real-time blood bank inventory.
6	The system determines whether sufficient compatible blood is available.
7	If donor support is required, the system identifies compatible eligible donors within the specified 10 km radius.
8	The system sends emergency SMS alerts to the matched eligible donors.
9	The system displays confirmation of the request and records the notification status.
10	Use case ends successfully.

Alternate Flow

6a. Insufficient Blood Inventory	The required blood component is unavailable or the available quantity is insufficient.
6a1.	The system initiates compatible donor matching.
6a2.	The system filters eligible, compatible donors within the specified 10 km radius.
6a3.	The system sends emergency SMS alerts to the matched donors.
6a4.	The system records the notification status and informs the Emergency Requester that donor support has been initiated.